

Social Networks and Social Capital (SOCI 3102)

Tutorial 3: QIN Xuanlong

Agenda

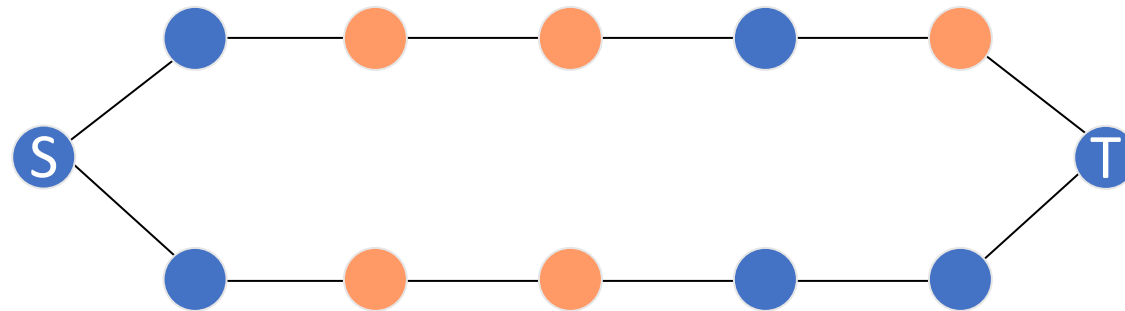
- Recap
 - Segmentation and Cohesion - James Moody
- R Markdown
- R Tutorial
 - Clustering Algorithms

Tutorial Materials

- Tutorial3_recap.html
- Tutorial3.html
- Dataset
 - Amherst_College.csv – group 1
 - Brown_University.csv – group 2
 - Florida_State_Univetsity.csv – group 3
 - Assignment 3-random grouping.xlsx

Recap – Social cohesion and connectivity

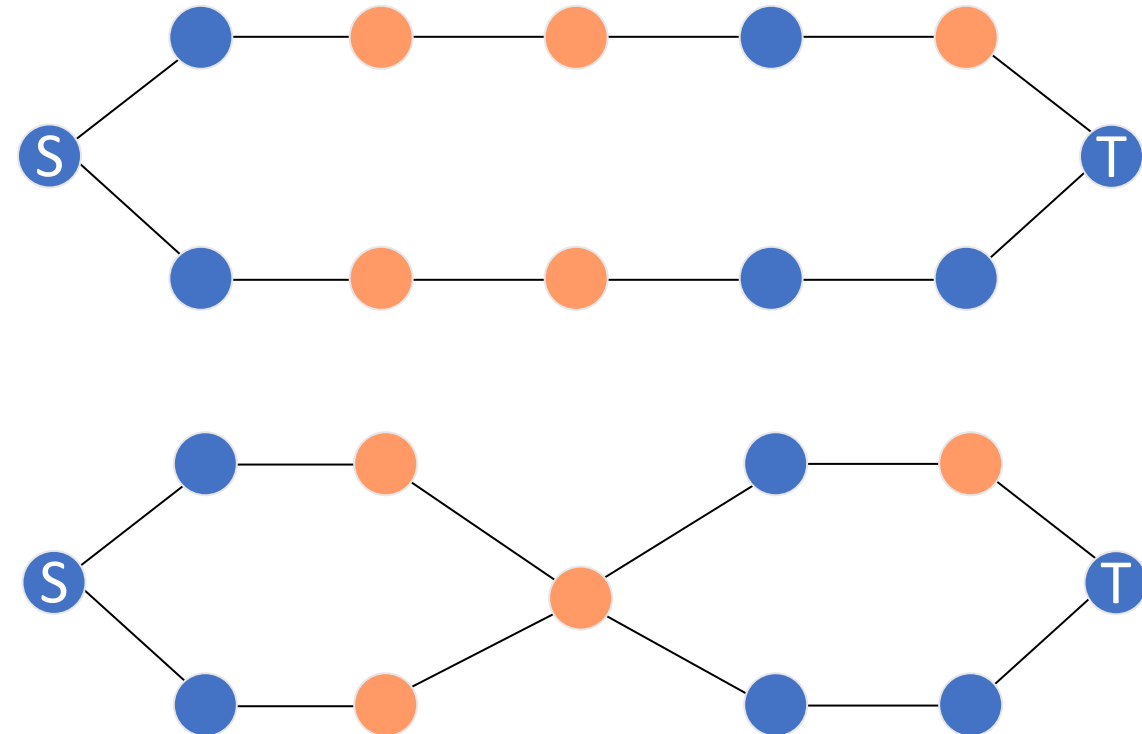
- Diffusion implication of relational structure
- Considering the following scenario:
 - Probability that actor i infects actor j (p_{ij}) is a constant over all relations = 0.6
 - S & T are connected through the following structure:
 - The probability that S infects T through either path would be: [0.09](#)



- $P = 0.6 * 0.6 * 0.6 * 0.6 * 0.6 * 0.6 + 0.6 * 0.6 * 0.6 * 0.6 * 0.6 * 0.6 = 0.09$

However...

- Every actor but one has the exact same number of partners
- The category-to-category (blue to orange) mixing is identical
- The distance from S to T is the same (7 steps)
- S and T have not changed their behavior
- Their partner's partners have the same behavior
- But the probability of an infection moving from S to T is: $= 0.186$
- Different structures create different outcomes



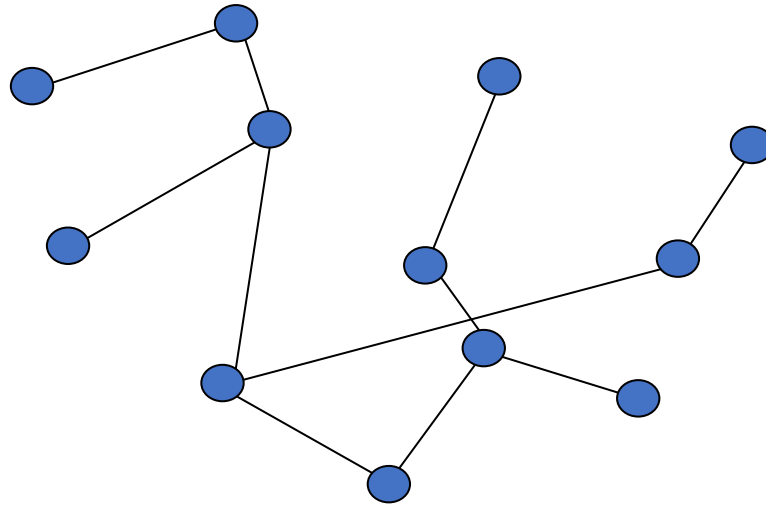
- Structural Cohension

- “Instead of focusing on the local involvement of actors (that their friends are friends with each other, or that there are a few with many ties), we ask what relational configuration is required for a cohesive network? we focus directly on the minimum requirements for holding a network together ” - James Moody(2003, PAA)

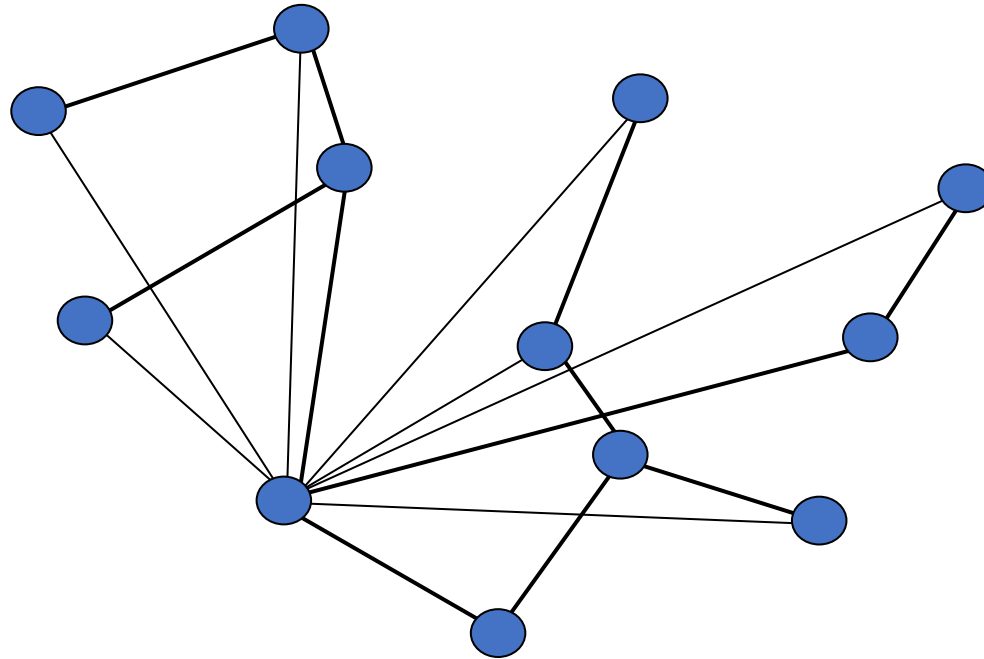
- An intuitive definition of structural cohesion:

- A collectivity is structurally cohesive to the extent that the social relations of its members hold it together.

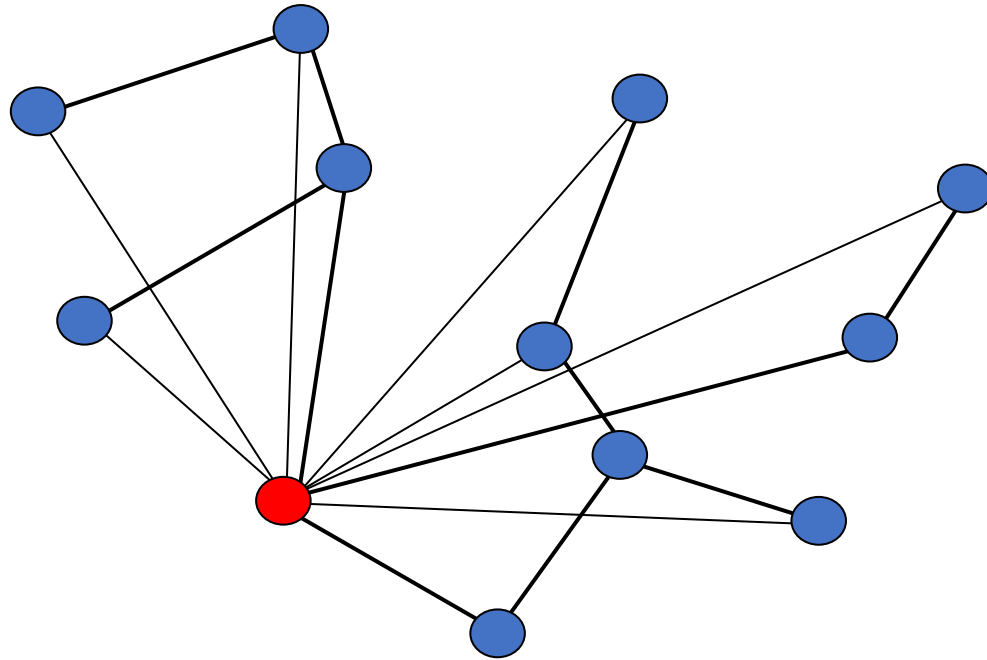
- The minimum requirement for structural cohesion is that the collection be connected.



- Now imaging we increase relational volume,

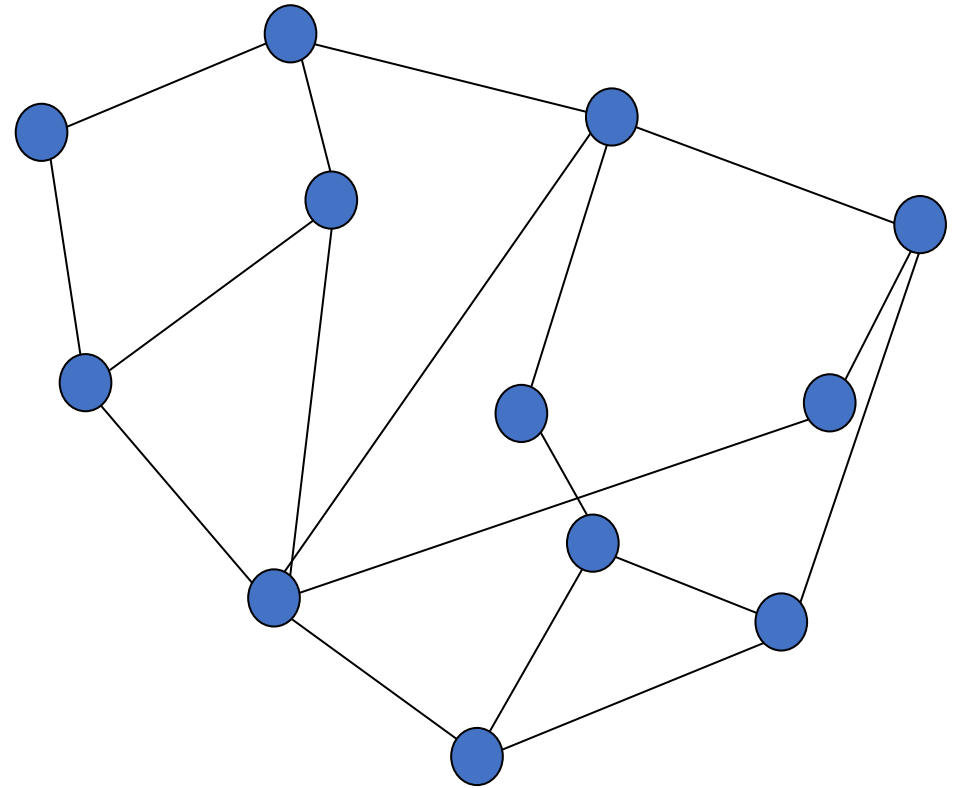


- When focused on one node, the system is still fragile. Why?

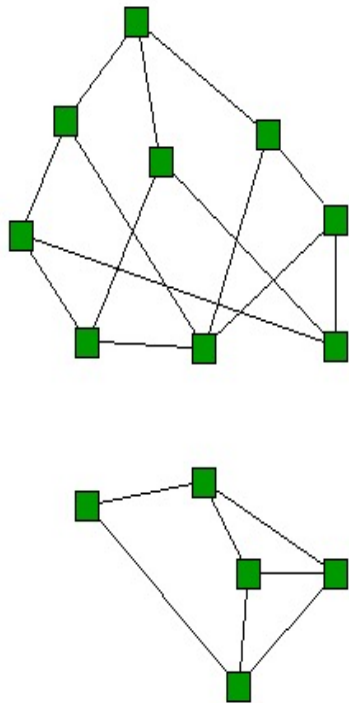


- The system's connectivity depends crucially on a single person.

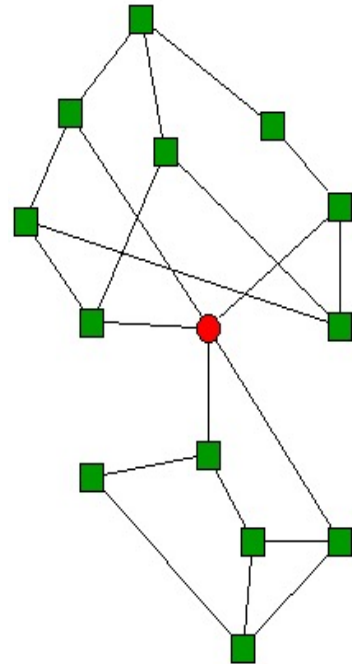
- If, on the other hand, the same number of ties are distributed widely throughout the network, many cycles form that re-connect the network, and no single star becomes the sole connector for the network.
- Spreading relations around the structure make it robust to node removal.



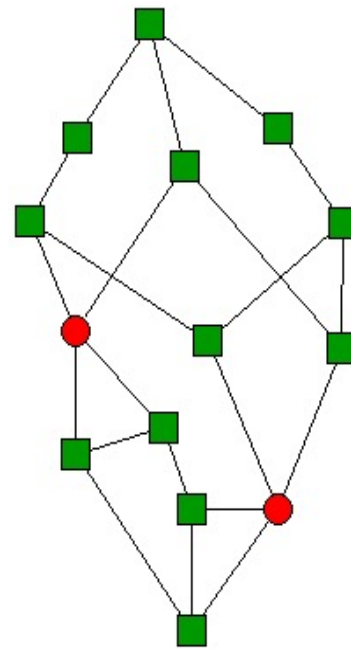
- Networks are structurally cohesive if they remain connected even when nodes are removed



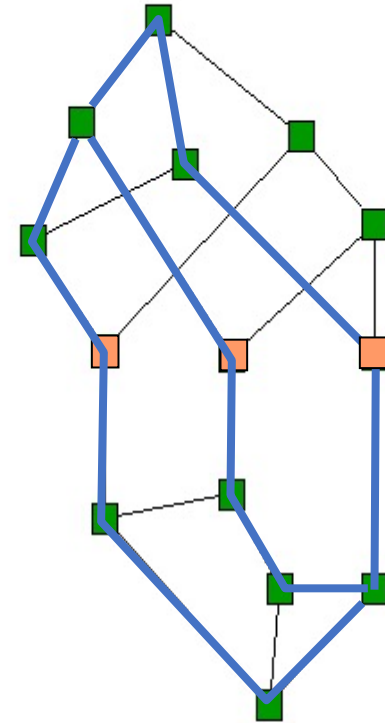
0



1

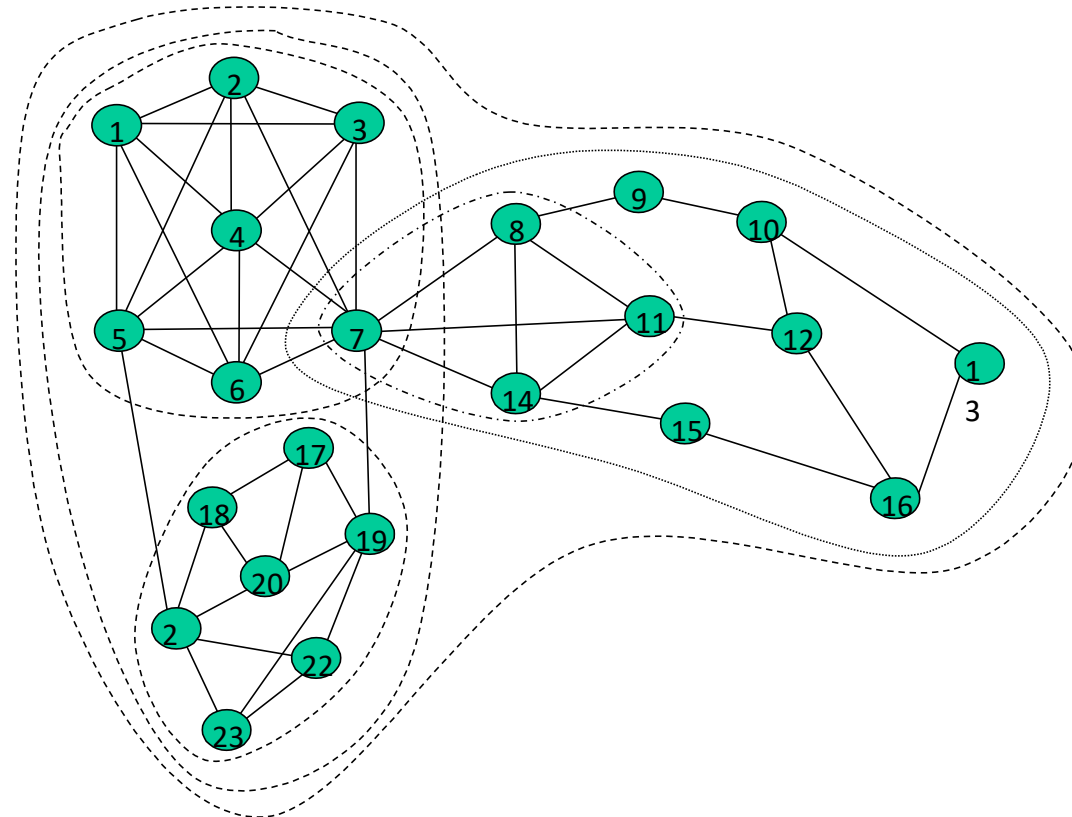


2



3

- Structural cohesion gives rise automatically to a clear notion of embeddedness, since cohesive sets nest inside of each other.



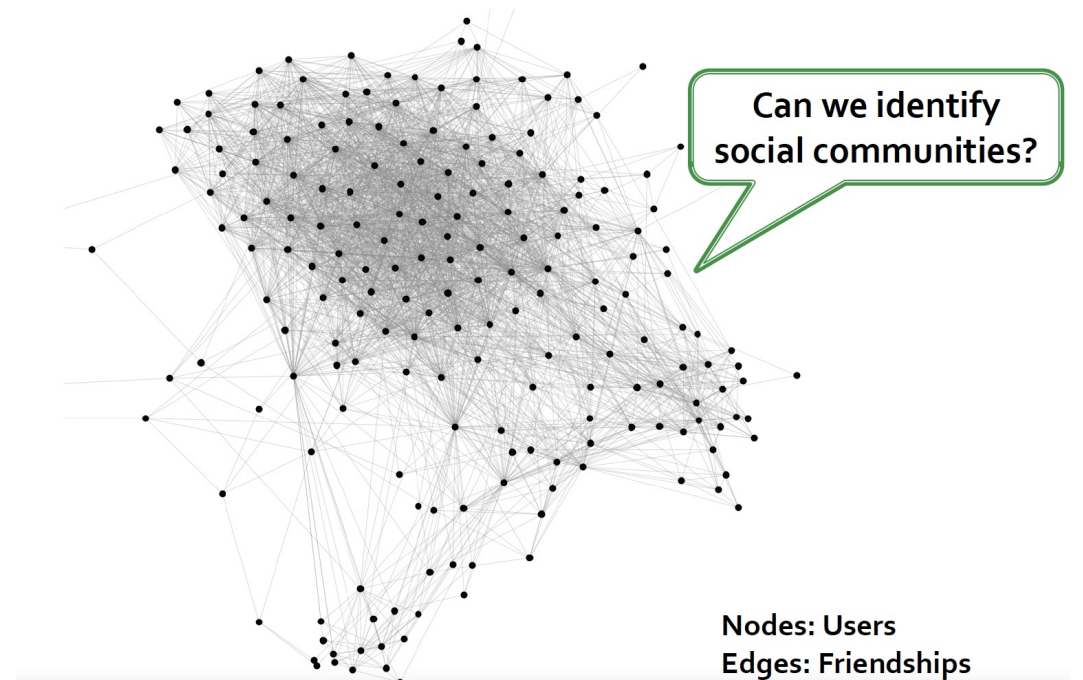
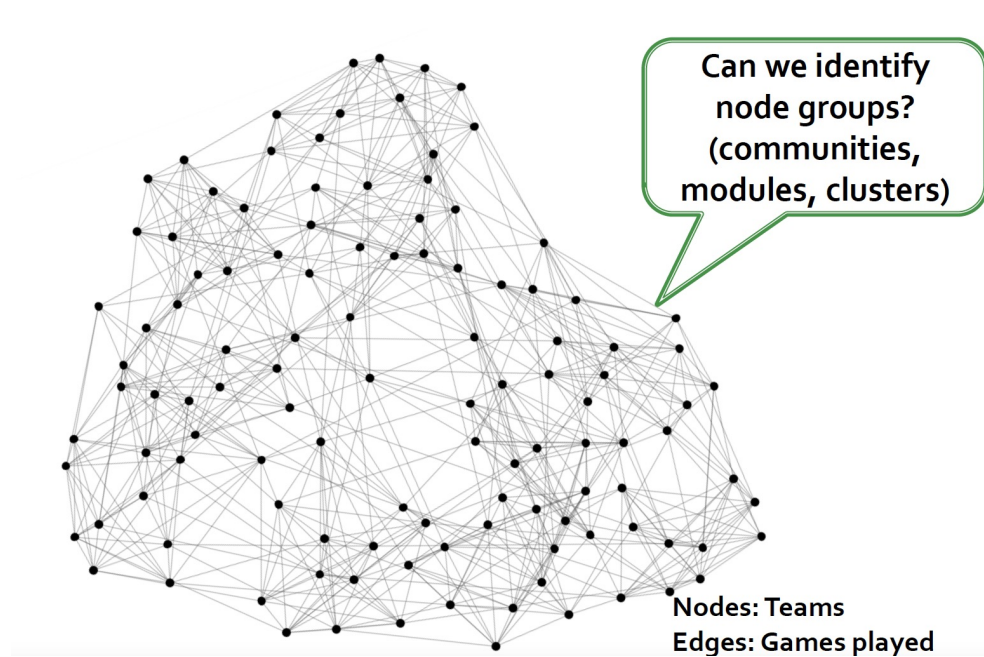
- There is a strong connection between the literature on “Social Capital” and group membership, which provides a theoretical link between notions of structural cohesion, ideational diffusion, and the duality of groups.
- And we will learn how to conduct community detecting using clustering algorithms.

Community Detection

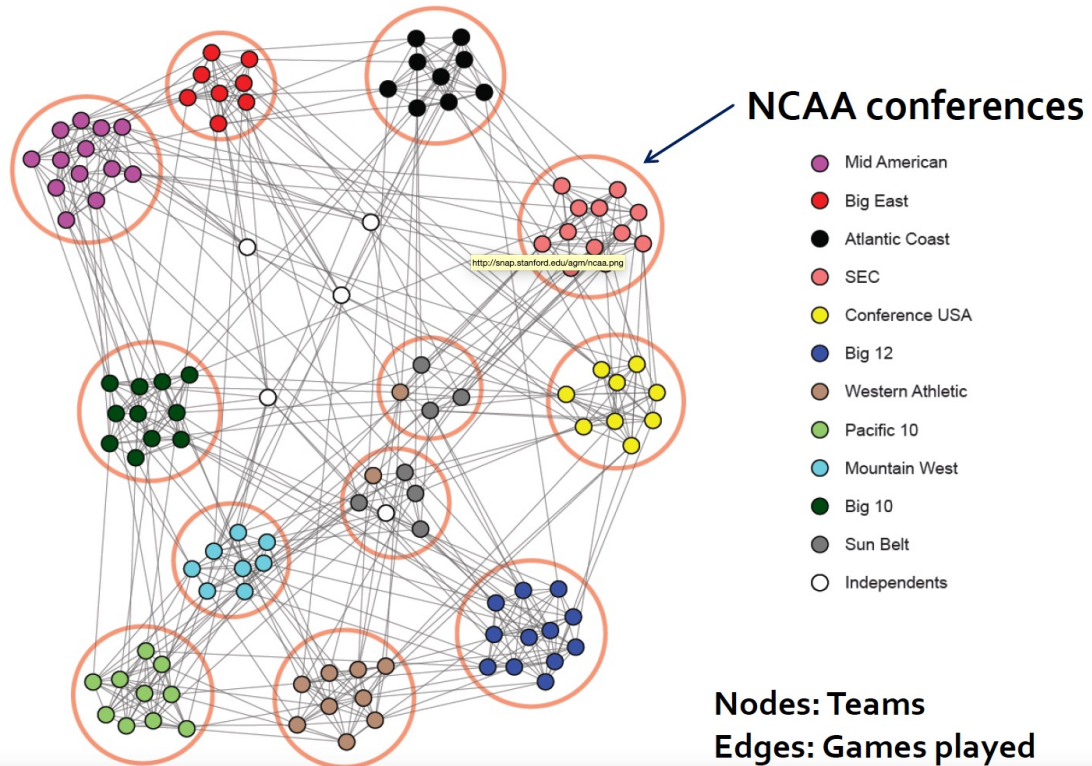
How to find communities?

- How to automatically find such densely connected groups of nodes?
- Ideally such automatically detected clusters would then correspond to real groups.

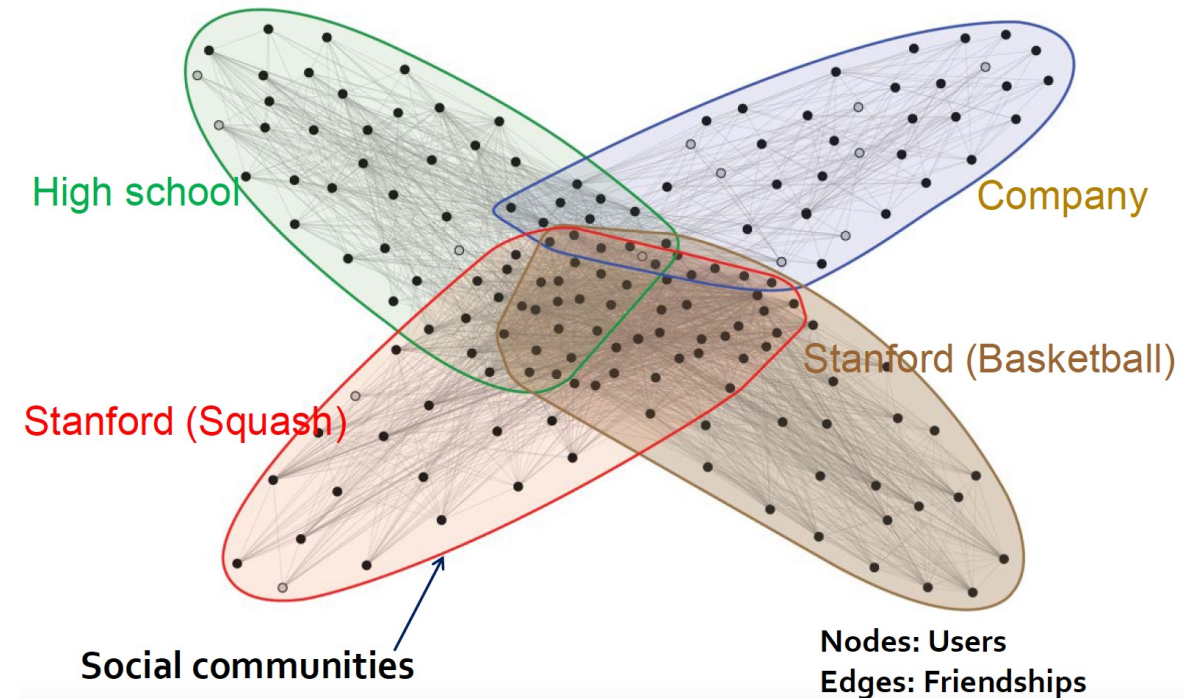
- Actually, It is hard for us to find communities, modules, clusters by eyes.



NCCA Football Networks



Facebook Ego-network



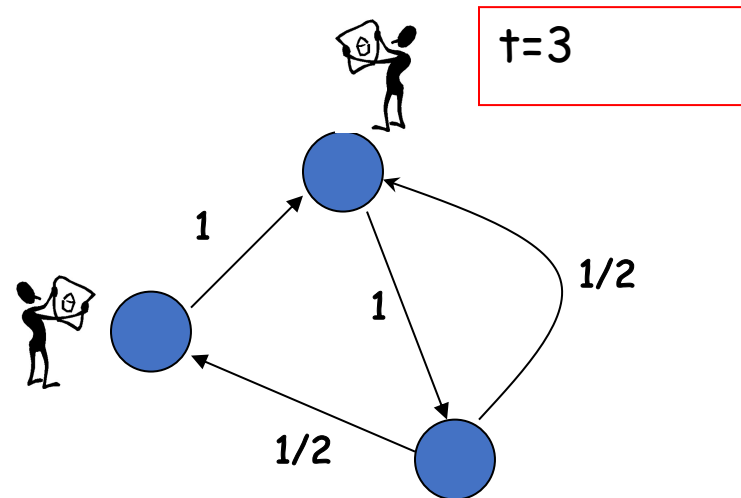
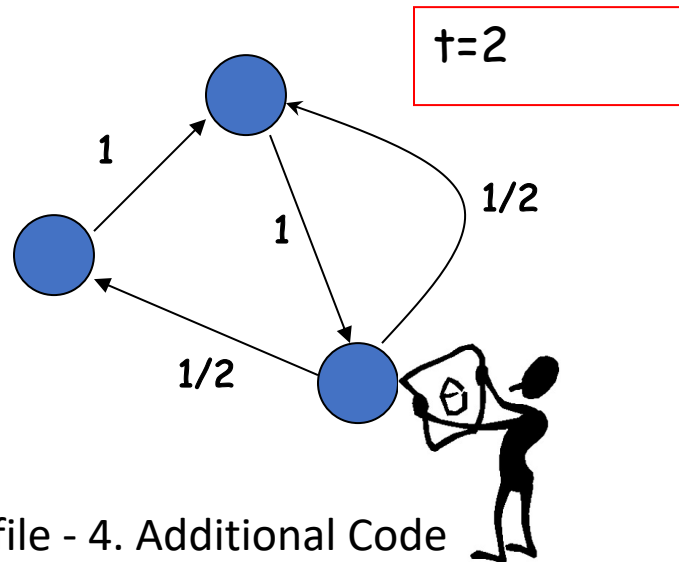
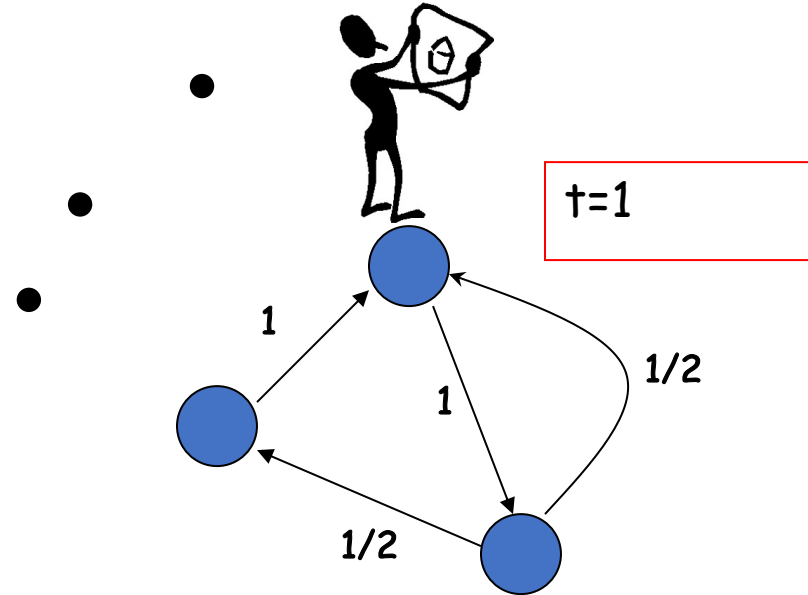
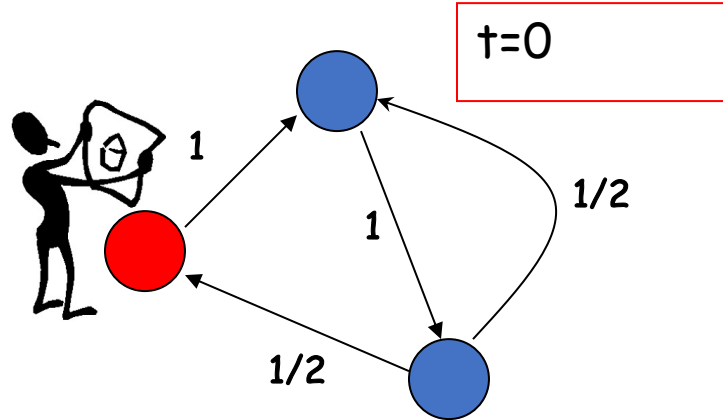
Dataset on Graph

- Then,
 - We would like to borrow detection algorithms to get the communities.
- The underlying data is naturally a graph
 - Papers linked by citation
 - Authors linked by co-authorship
 - Bipartite graph of customers and products
 - Web-graph
 - Friendship networks: who knows whom
- I will briefly introduce **random walk** and infomap algorithms without mathematic inference.

Random walk

- The principle behind a random walk is quite simple: you start at a given node and then move to a neighboring node at random. This process is repeated for many steps.
- In the context of social networks, a random walk can be used to measure the "closeness" of nodes based on how often a random walker visits them.
- The intuition is that if many random walks starting from different nodes frequently visit a certain group of nodes, these nodes are more likely to form a community.

What is a random walk



Infomap

- The notion of "random walks" is not an algorithm by itself but a component that can be used in various algorithms.
- The Infomap algorithm uses random walks in a more specific way. It is **based on the idea of information flow** and uses the concept of random walks to uncover community structures.
- The algorithm uses the theory of information **to find the most efficient encoding of the random walk path**. In practice, this means that the algorithm seeks a partition of the network into clusters (communities) such that the description of the random walk trajectory is as short as possible.

Community Detection Algorithms

- Some famous detection algorithms has been packed in “igraph” package.
 - Louvain
 - Leiden
 - Walktrap
 - Infomap
 - Newman
- You could find more details at <https://igraph.org/c/doc/igraph-Community.html>

R Manual for SOCI 3102

More on `igraph` package

The main purpose is that you can write codes for analyzing a given network

Agenda

- R Markdown
- Recap tutorial 1 & 2
- Tutorial 3: More on igraph
 - Calculate Network density
 - Infomap & random walk
 - Personalize your graph
- Introduction of Problem Set 3

R Markdown

Before we get started...

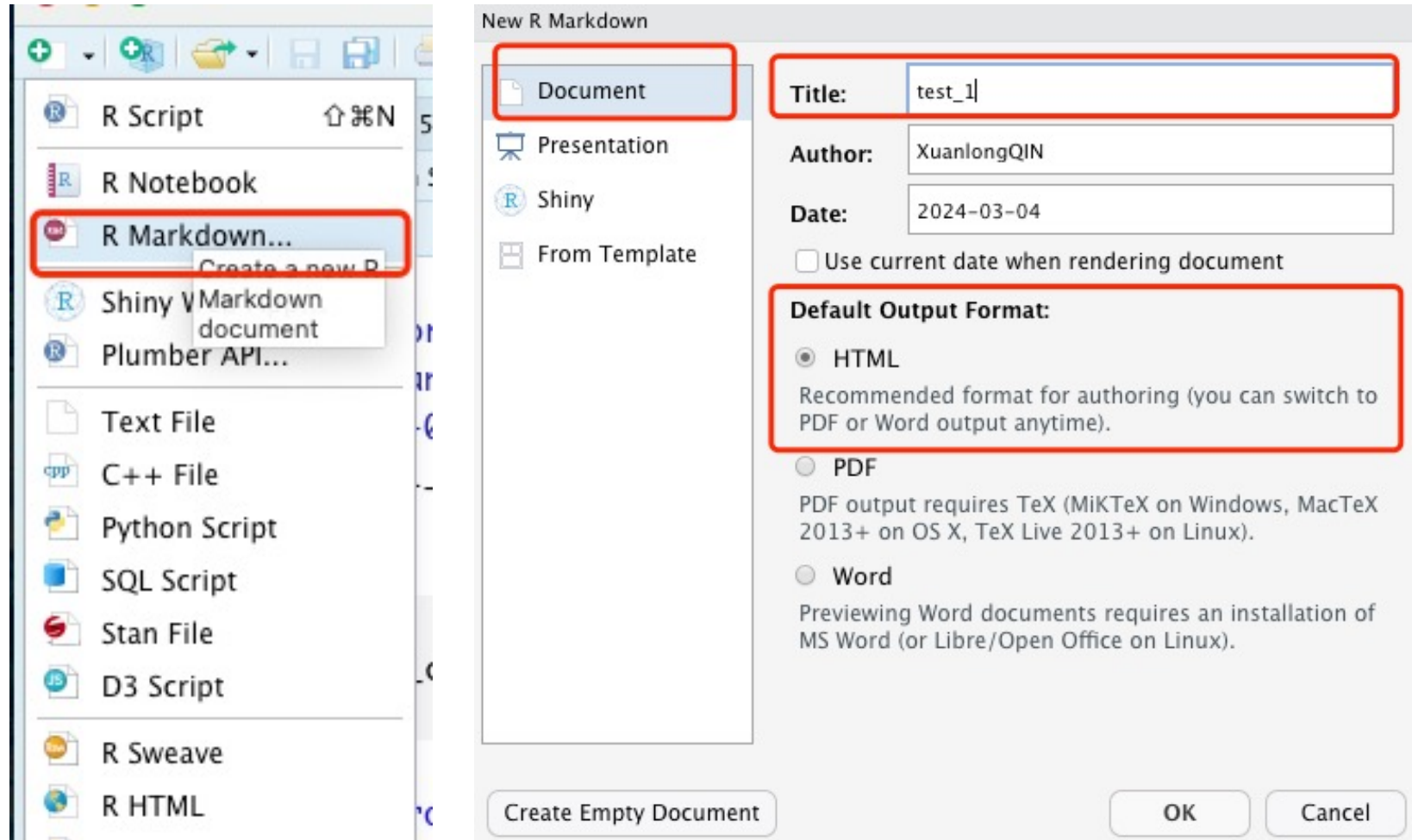
The third assignment needs to be finalized by R Markdown.

- We only receive *.Rmd + *.html file in Blackboard.
- Please open Rstudio and be ready for practice.

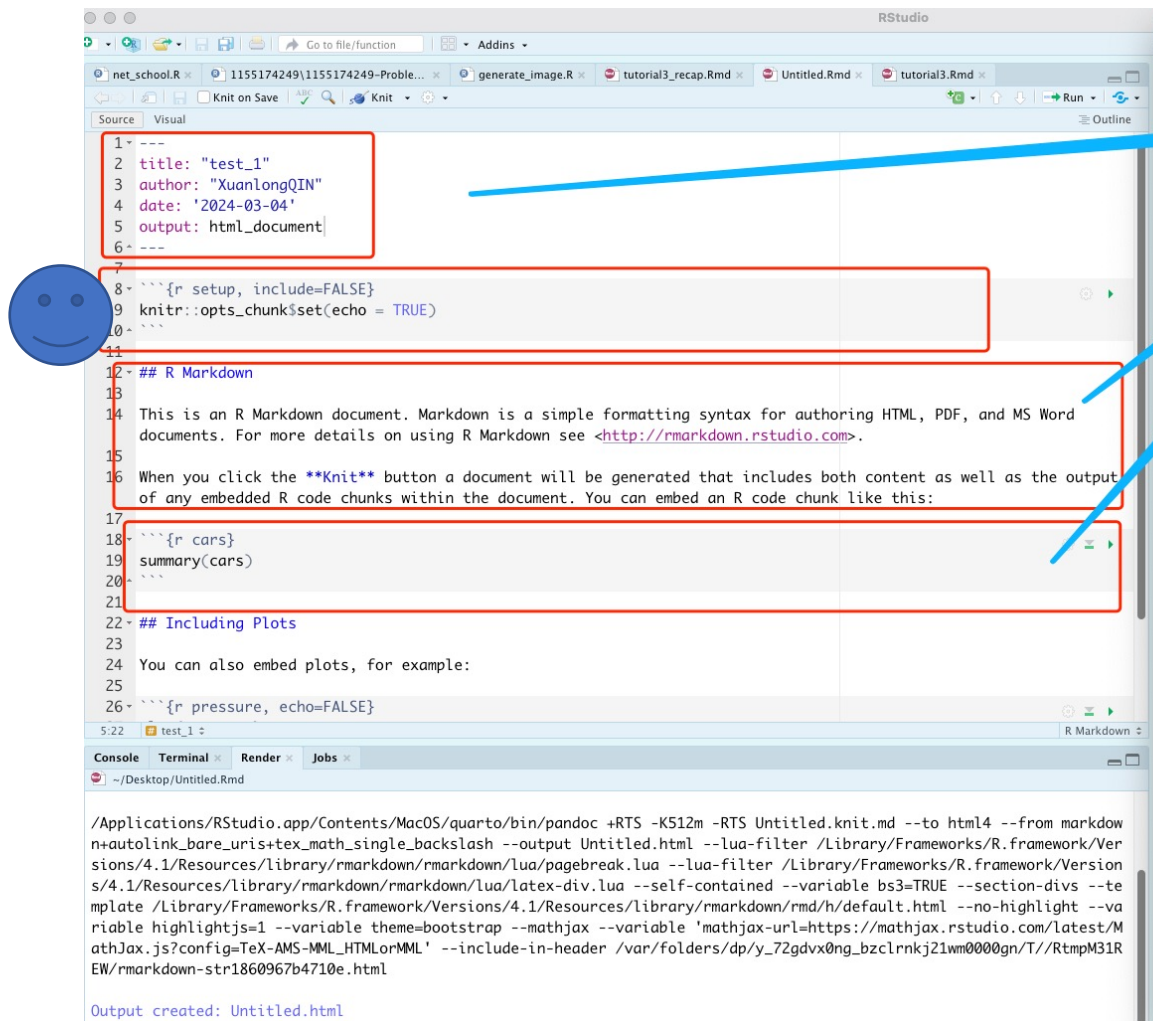
What is R Markdown

- R Markdown documents are fully reproducible. Use a productive notebook interface to weave together narrative **text** and **code** to produce elegantly formatted output. Use multiple languages including R, Python, and SQL.
- Official website: <https://rmarkdown.rstudio.com/>
- `install.packages("rmarkdown")`

New a markdown script



Knit and Save your file



```
1 ---
2 title: "test_1"
3 author: "XuanlongQIN"
4 date: '2024-03-04'
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.
15
16 When you click the Knit button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:
17
18 ```{r cars}
19 summary(cars)
20 ```
21
22 ## Including Plots
23
24 You can also embed plots, for example:
25
26 ```{r pressure, echo=FALSE}
```

5:22 test_1

Console Terminal Render Jobs

~/Desktop/Untitled.Rmd

/Applications/RStudio.app/Contents/MacOS/quarto/bin/pandoc +RTS -K512m -RTS Untitled.knit.md --to html4 --from markdown+autolink_bare_uris+tex_math_single_backslash --output Untitled.html --lua-filter /Library/Frameworks/R.framework/Versions/4.1/Resources/library/rmarkdown/rmarkdown/latex-div.lua --lua-filter /Library/Frameworks/R.framework/Versions/4.1/Resources/library/rmarkdown/rmarkdown/latex-div.lua --self-contained --variable bs3=TRUE --section-divs --template /Library/Frameworks/R.framework/Versions/4.1/Resources/library/rmarkdown/rmd/h/default.html --no-highlight --variable highlightjs=1 --variable theme=bootstrap --mathjax --variable 'mathjax-url=https://mathjax.rstudio.com/latest/MathJax.js?config=TeX-AMS-MML_HTMLorMML' --include-in-header /var/folders/dp/y_72gdvx0ng_bzclrnkj21wm0000gn/T/RtmpM31REW/rmarkdown-str1860967b4710e.html

Output created: Untitled.html

test_1

XuanlongQIN

2024-03-04

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

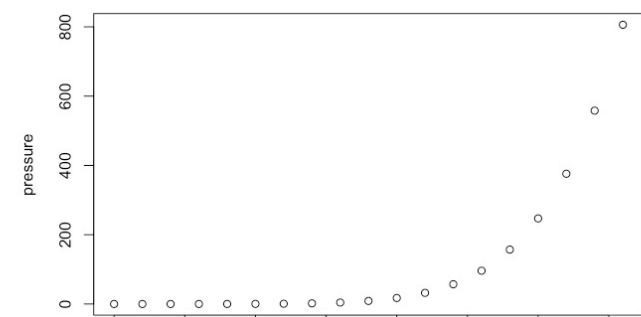
```
summary(cars)
```

	speed	dist
## Min.	: 4.0	Min. : 2.00
## 1st Qu.	:12.0	1st Qu.: 26.00
## Median	:15.0	Median : 36.00
## Mean	:15.4	Mean : 42.98
## 3rd Qu.	:19.0	3rd Qu.: 56.00
## Max.	:25.0	Max. :120.00



Including Plots

You can also embed plots, for example:



Chunk Options

Chunk output can be customized with [knitr options](#), arguments set in the {} of a chunk header. Above, we use five arguments:

- `include = FALSE` prevents code and results from appearing in the finished file. R Markdown still runs the code in the chunk, and the results can be used by other chunks.
 - `echo = FALSE` prevents code, but not the results from appearing in the finished file. This is a useful way to embed figures.
 - `message = FALSE` prevents messages that are generated by code from appearing in the finished file.
 - `warning = FALSE` prevents warnings that are generated by code from appearing in the finished.
 - `fig.cap = "..."` adds a caption to graphical results.
- See the [R Markdown Reference Guide](#) for a complete list of knitr chunk options.
 - ***Generally, I won't compulsorily ask you to change the default setting, and in our tutorial, you could ignore these code chunks.***

Basic Symbol

- Bash --- for header
- Backquote ` for R code chunks
- Parentheses {} for code formatting
- Number # for annotations



Recap: Tutorials Overview

Based on R Markdown

- Tutorial 1,
 - Install R, R Studio, install certain network packages, review and application of basic network concepts
- Tutorial 2,
 - Basic & data manipulation in R, building network graphs
- **Tutorial 3.**
 - **More on network graphs and basic network statistics in R**

Basic concepts

- Data type
 - Vector, Matrix, Array, Data Frame, Lists
- View Dataset
- Data descriptions
- Input external data
- Help function
- Please open tutorial3_recap.rmd file by RStudio

Basic `igraph` package

- Install igraph
 - Create simple graph
 - Indicators of Networks
 - How to personalized your network
-
- Please open tutorial3_recap.rmd file by Rstudio

Tutorial 3: More on igraph

Community detection

Community detection

- 1. Check and clean data
 - 2. Calculate Network Density
 - 3. Conducting Clustering Algorithms (Take walktrap as an example)
 - 4. Define the colors for different communities
-
- Please open tutorial3.Rmd file by Rstudio

About Problem Set 3

Datasets - Academic citation network

- The datasets include the fields of articles published by the university in 2011 and the fields of these papers' references.
 - E.g. "Information flow through strong and weak ties in intraorganizational social networks", "Social Science", "Medical Innovation: A Diffusion Study", "Medicine"
 - Then, "Social Science", "Medicine" were recorded.
- I collected three universities' published papers in 2011 as our datasets.
- Three universities (Amherst College, Brown University, and Florida State University)
 - Liberal and arts, Science
 - Status
 - Three datasets will be randomly assigned to each student.

Data description

- Data description
 - Each dataset have two columns
 - Original: a paper from a specific field
 - Destination: the field of cite paper
 - For instance, A, Computer Science, B, Sociology.
 - A Cited B
 - a directed network

Notes:

- 1. you should set a fix random seed.
 - `set.seed(123)`
- 2. Please check whether the code I gave is available.
 - Ensuring the Return value is true !!!
- 3. Everyone must submit their assignment by *.Rmd format.
 - Prefer “student ID_name_group.Rmd”
 - i.e. 1155186031_xuanlong_group1.Rmd
- 4. You have extra one week for finish this assignment.
 - Due: 28th, Mar.